

Course Instructor:

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Reference Books:

- Linear Algebra with Supplemented Applications by Howard Anton/ Chris Rorres, Edition 10.
- Introductory Linear Algebra with Applications by Bernard Kolman, David R. Hill, Edition 9.
- Linear Algebra with applications by Otto Brestscher.

Assessments

Quizzes (5)	→	15%
Assignments (4)	→	10%
Class Participation	→	10%
Mid Term	→	20%
Final Term	→	45%

What do you think about Linear Algebra?

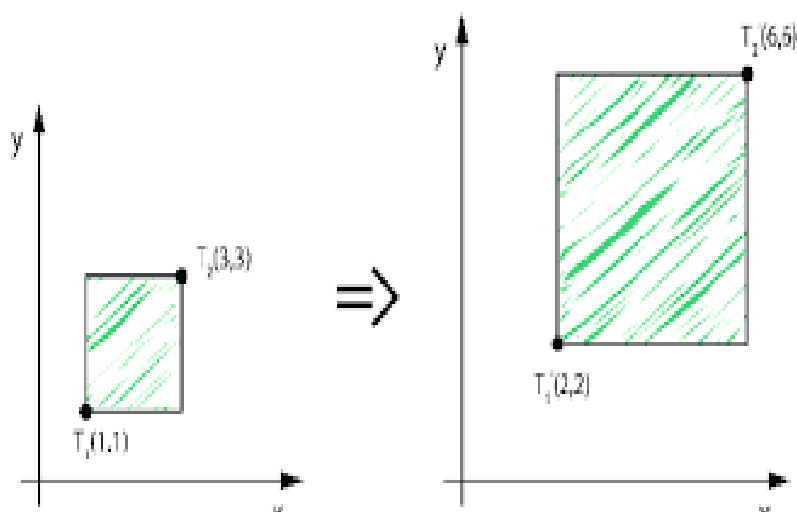
- Algebra is the art of solving equations and systems of equations.
- Linear algebra, then, is the art of solving system of linear equations.

Why We Study Linear Algebra in Computer Science?

When you take a digital photo with your phone or transform the image in Photoshop, when you play a video game or watch a movie with digital effects, when you do a web search or make a phone call, you are using technologies that build upon linear algebra.

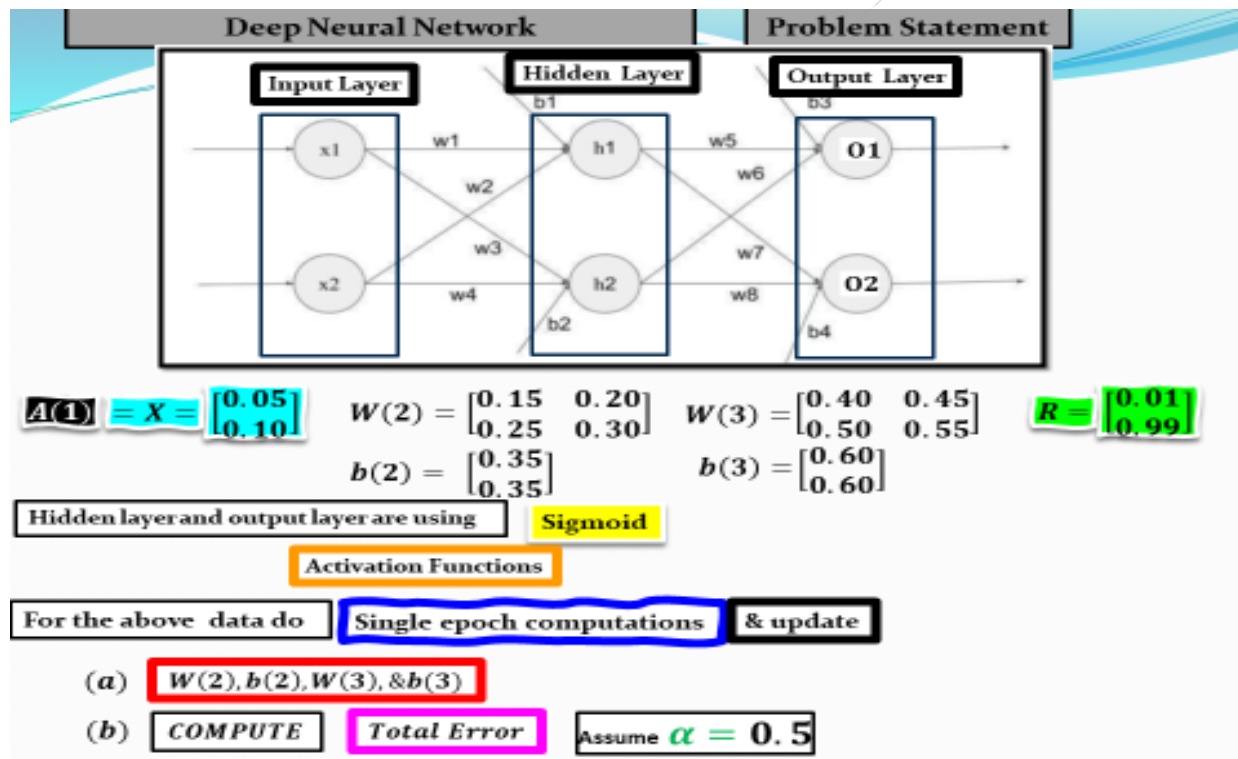
Linear algebra provides concepts that are crucial to many areas of computer science, including graphics, image processing, cryptography, machine learning, computer vision, optimization, graph algorithms, quantum computation, computational biology,

information retrieval and web search. Linear algebra in turn is built on two basic elements, the matrix and the vector.





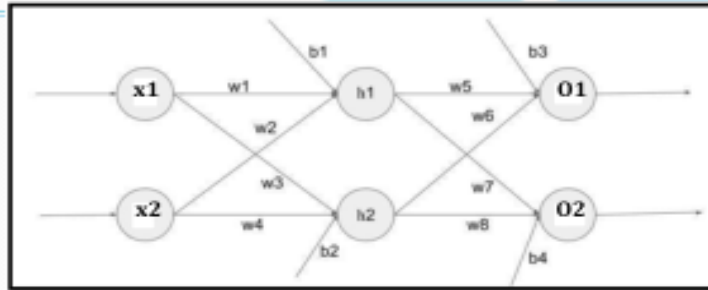
Deep Learning (Neural Networks)



Deep Neural Network

Forward Pass

Computations



$$A(1) = X = \begin{bmatrix} 0.05 \\ 0.10 \end{bmatrix} \quad W(2) = \begin{bmatrix} 0.15 & 0.20 \\ 0.25 & 0.30 \end{bmatrix} \quad W(3) = \begin{bmatrix} 0.40 & 0.45 \\ 0.50 & 0.55 \end{bmatrix} \quad R = \begin{bmatrix} 0.01 \\ 0.99 \end{bmatrix}$$

$$B(2) = \begin{bmatrix} 0.35 \\ 0.35 \end{bmatrix} \quad B(3) = \begin{bmatrix} 0.60 \\ 0.60 \end{bmatrix}$$

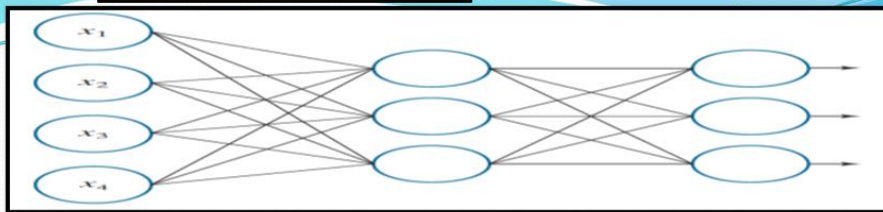
$$Z(2) = W(2)A(1) + B(2) = \begin{bmatrix} 0.15 & 0.20 \\ 0.25 & 0.30 \end{bmatrix} \begin{bmatrix} 0.05 \\ 0.10 \end{bmatrix} + \begin{bmatrix} 0.35 \\ 0.35 \end{bmatrix} = \begin{bmatrix} 0.3775 \\ 0.3925 \end{bmatrix}$$

$$Z(3) = W(3)A(2) + B(3) = \begin{bmatrix} 0.40 & 0.45 \\ 0.50 & 0.55 \end{bmatrix} \begin{bmatrix} 0.5933 \\ 0.5969 \end{bmatrix} + \begin{bmatrix} 0.60 \\ 0.60 \end{bmatrix} = \begin{bmatrix} 1.1059 \\ 1.2249 \end{bmatrix}$$

$$A(2) = \sigma(Z(2)) = \begin{bmatrix} \frac{1}{1 + e^{-0.3775}} \\ \frac{1}{1 + e^{-0.3925}} \end{bmatrix} = \begin{bmatrix} 0.5933 \\ 0.5969 \end{bmatrix} \quad A(3) = \sigma(Z(3)) = \begin{bmatrix} \frac{1}{1 + e^{-1.1059}} \\ \frac{1}{1 + e^{-1.2249}} \end{bmatrix} = \begin{bmatrix} 0.7513 \\ 0.7729 \end{bmatrix}$$

Deep Neural Network

Example No. 2



$$W(2) =$$

$$b(2) =$$

$$W(2) = \begin{bmatrix} W_{11} & W_{12} & W_{13} & W_{14} \\ W_{21} & W_{22} & W_{23} & W_{24} \\ W_{31} & W_{32} & W_{33} & W_{34} \end{bmatrix}$$

$$b(2) = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$W(3) =$$

$$b(3) =$$

$$W(3) = \begin{bmatrix} W_{11} & W_{12} & W_{13} \\ W_{21} & W_{22} & W_{23} \\ W_{31} & W_{32} & W_{33} \end{bmatrix}$$

$$b(3) = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Total Parameters
Trained in single epoch

Weight Parameters : 21
(3 x 4, 3 x 3)

Bias Parameters : 6
(3 x 1, 3 x 1)

Activate Window
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Linear Equation:

An equation whose **exponent** or **power** is **one** is called **linear equation**.

Note:

- Linear equation does not involve any products of variables or roots of variables.
- All variables occur only to the first power and do not appear as arguments of trigonometric, logarithmic, or exponential functions.

Examples:

Equation	Type
$2x + 1 = 0$	Linear equation in One Variable
$x + 3y = 7$	Linear equation in Two Variables
$\frac{1}{2}x - y + 3z = -1$	Linear equation in Three Variables
$x_1 - 2x_2 - 3x_3 + x_4 = \ln(2)$	Linear equation in Four Variables
$x_1 + x_2 + x_3 + \cdots + x_n = 1$	Linear equation in n Variables

Linear Equation in General:

The equation

$$a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n = b \text{ --- (1)}$$

which expresses the real quantity **b** in terms of the unknowns **$x_1, x_2, x_3, \dots, x_n$** and the real constants **$a_1, a_2, a_3, \dots, a_n$** , is called a **linear equation**.

Question. Identify the following equations are **linear or nonlinear**? If it is **nonlinear**, which **term** in each equation is making it **nonlinear**?

1. $x + 3y^2 = 4$

2. $3x + 2y - xy = 5$

3. $\sin(x) + y = 0$

4. $\sqrt{x_1} + x_2 + x_3 = 1$

5. $3x + 2y = \cos(2)$

Exercise 1.1 [*Elementary Linear Algebra with Applications by Howard Anton*]

1. In each part, determine whether the equation is linear in x_1, x_2 and x_3 .

a) $x_1 + 5x_2 - \sqrt{2x_3} = 1$

b) $x_1 + 3x_2 + x_1x_3 = 2$

c) $x_1^{-2} + x_2 + 8x_3 = 5$

d) $x_1 = -7x_2 + 3x_3$

e) $x_1^{\frac{3}{5}} - 2x_2 + x_3 = 4$

f) $\pi x_1 - \sqrt{2x_2} + \frac{1}{3}x_3 = 7^{1/3}$

Solution of a Linear Equation

A **solution** to linear equation

$$a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n = b \text{ --- (1)}$$

is a sequence of n numbers $s_1, s_2, \dots, s_n$ which has the property that equation (1) is satisfied when $x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$ are substituted in equation (1).

For example: The equation $6x_1 - 3x_2 + 4x_3 = -13$
has the solution

$$x_1 = 2, \quad x_2 = 3, \quad x_3 = -4$$

Because on substituting these values, equation becomes identity i.e.;

$$6(2) - 3(3) + 4(-4) = -13$$
$$\text{or } -13 = -13$$

System of Linear Equation in two Variables

A system of linear equations in two variables x and y will have the form

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

Here a_i, b_i, c_i ($i = 1, 2$) are real numbers.

❖ To find a **solution to a linear system**, we already know two techniques called the

1. **method of elimination**
2. **method of substitution**
3. **We are going to learn new systematic techniques for complex systems.**

Note: Solution from any of these methods will remain same.

Example 1:

Find the **solution** of the linear system by using **method of elimination**.

$$\begin{array}{rcl} 5x + y = 3 & - & - & - & - & (1) \\ 2x - y = 4 & - & - & - & - & (2) \end{array}$$

Solution:

Let, we want to eliminate y , so by adding both equations (1) and (2), we get:

$$7x = 7 \Rightarrow x = 1$$

Put value of $x = 1$ in equation (1), we have

$$5(1) + y = 3 \Rightarrow y = -2$$

So, the **given system** has the **solution** $(x, y) = (1, -2)$.

Example 2:

Find the **solution** of the linear system by using **method of elimination**.

$$x - 3y = -7 \quad \text{--- (1)}$$

$$2x - 6y = 7 \quad \text{--- (2)}$$

Solution

We want to eliminate x , so by multiplying equation (1) by “2” and subtracting from equation (2), we get

$$\begin{array}{r} 2x - 6y = -14 \\ -2x + 6y = -7 \\ \hline 0 = -21 \end{array}$$

which makes no sense. This means that the **given system** has **no solution**.

Practice Questions:

7. In each part, determine whether the given **point** is a **solution** of the **linear system** or not?

$$\begin{cases} 2x - 4y - z = 1 \\ x - 3y + z = 1 \\ 3x - 5y - 3z = 1 \end{cases}$$

- a) (3, 1, 1)
- b) (3, -1, 1)
- c) (13, 5, 2)
- d) $(\frac{13}{2}, \frac{5}{2}, 2)$
- e) (17, 7, 5)

8. In each part, determine whether the given point is a solution of the linear system

$$\begin{cases} x + 2y - 2z = 3 \\ 3x - y + z = 1 \\ -x + 5y - 5z = 5 \end{cases}$$

- a) $(\frac{5}{7}, \frac{8}{7}, 1)$
- b) $(\frac{5}{7}, \frac{8}{7}, 0)$
- c) (5, 8, 1)
- d) $(\frac{5}{7}, \frac{10}{7}, \frac{2}{7})$
- e) $(\frac{5}{7}, \frac{22}{7}, 2)$

11. In each part, find a system of linear equations corresponding to the given augmented matrix

a) $\begin{pmatrix} 2 & 0 & 0 \\ 3 & -4 & 0 \\ 0 & 1 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 3 & 0 & -2 & 5 \\ 7 & 1 & 4 & -3 \\ 0 & -2 & 1 & 7 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 & 0 & 0 & 7 \\ 1 & 2 & 4 & 0 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 1 & 0 & 0 & 0 & 7 \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 4 \end{pmatrix}$

Work to do:

Exercise 1.1 [*Elementary Linear Algebra with Applications* by **Howard Anton**]

Question 1 -10.